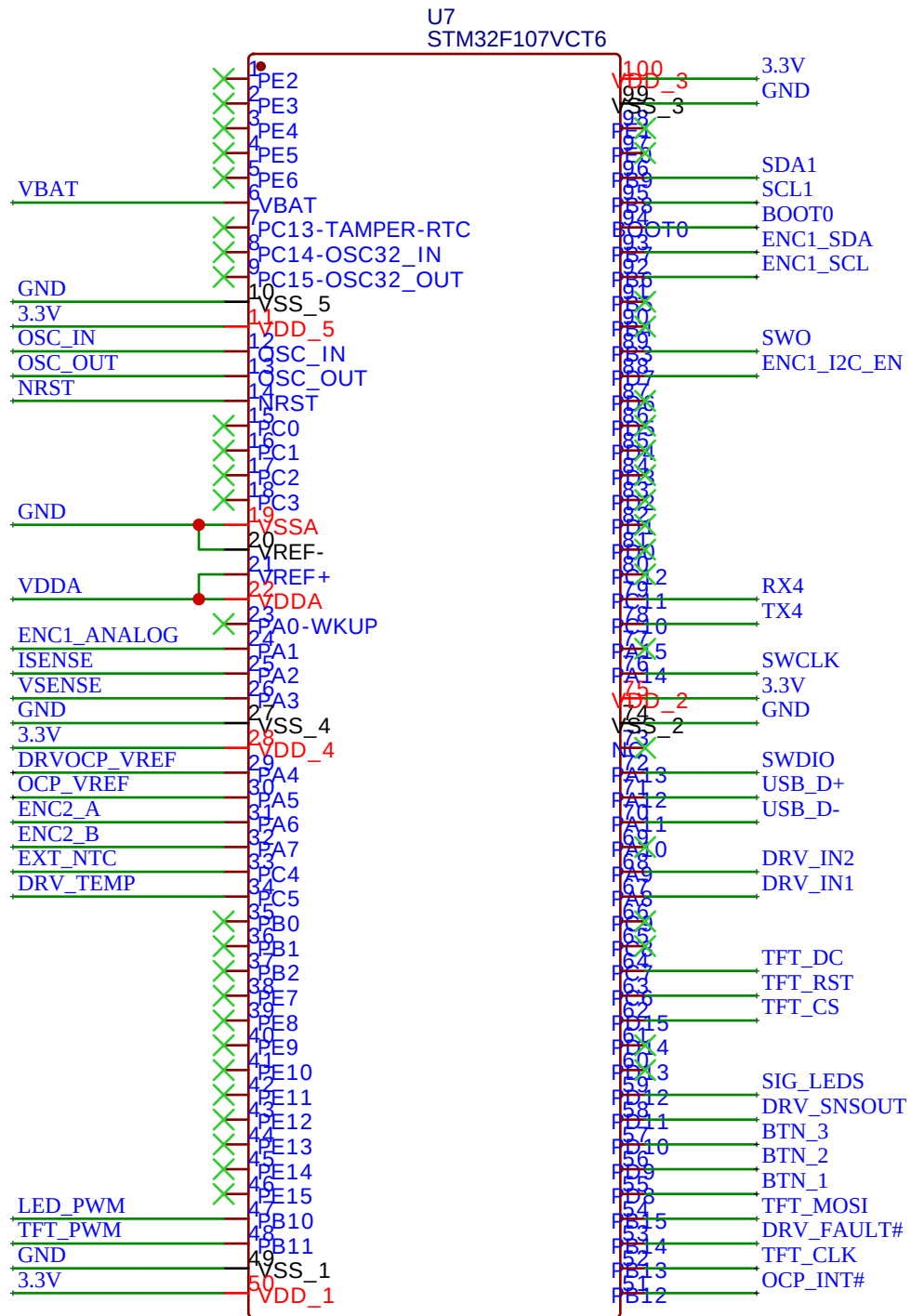
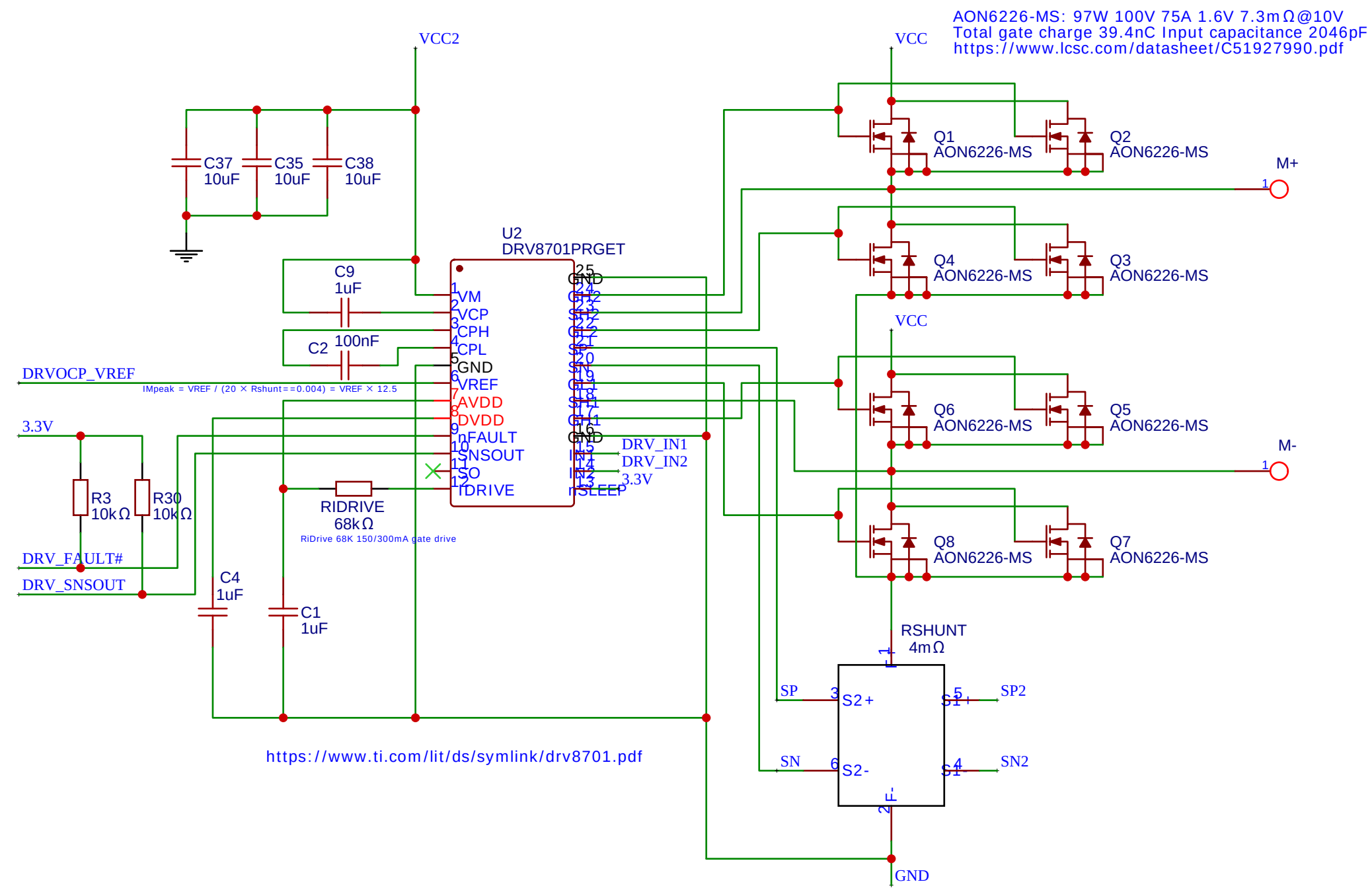


DC Motor Controller with Velocity and PWM Mode

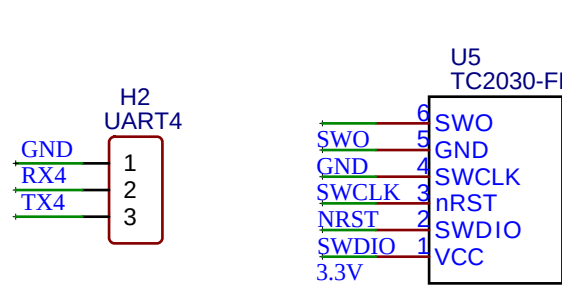
- 8-36V, 20A continuous, 40A peak
- Adjustable cycle by cycle motor current limit (0.5-40A)
- Adjustable max. current with signal LED (0.5-40A)
- 1.14" 135x240 TFT display, rotary encoder and 3 buttons for easy navigation
- Support for magnetic encoder MT6701 in A/B mode up to 1024PPR (55000 RPM)
- Dimmable LED with CC driver 3-32V/350mA/5W
- Current, voltage and temperature monitoring

Firmware: https://github.com/sascha432/motor_controller_stm32

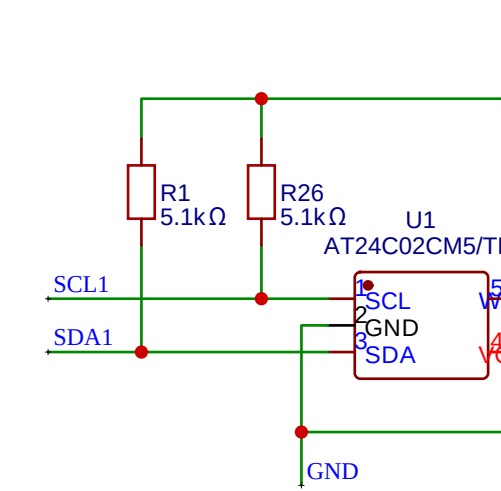
Motor driver and MOSFETS



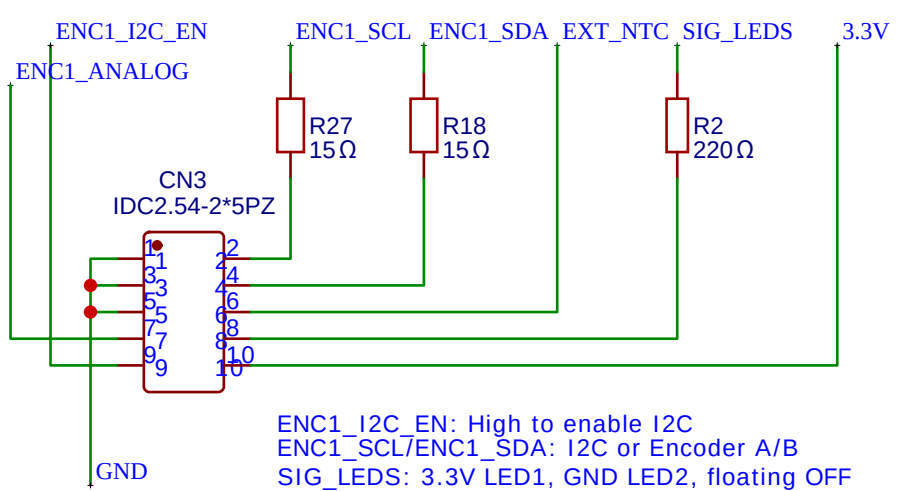
Programming/debug header



EEPROM and I2C pullups



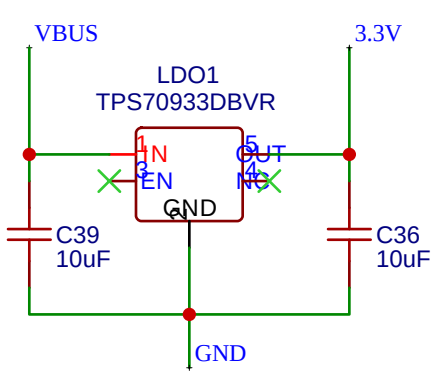
Externals & MT6701 connector



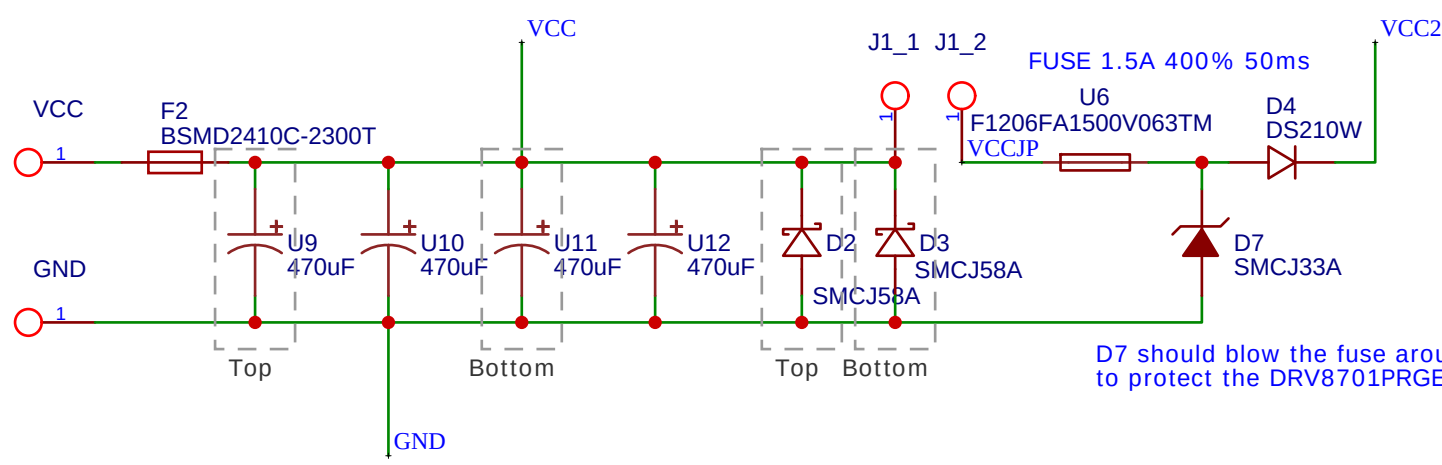
Input voltage detection up to 40V



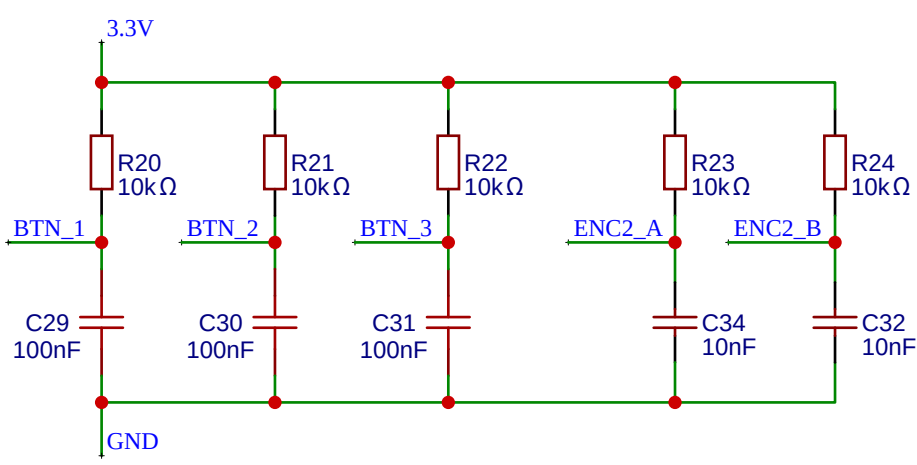
3.3V power supply with reverse current protection for the USB port (optional)



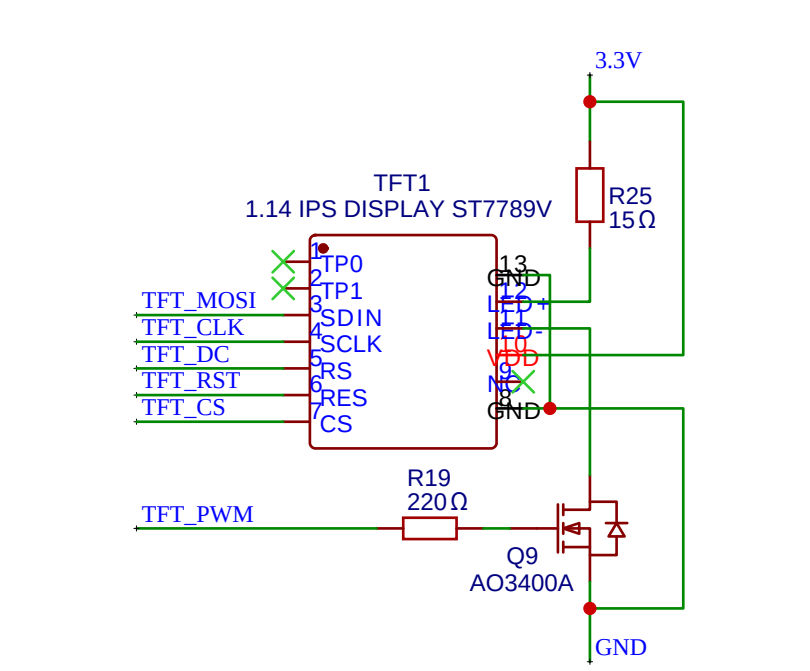
Input bulk capacitance and protection



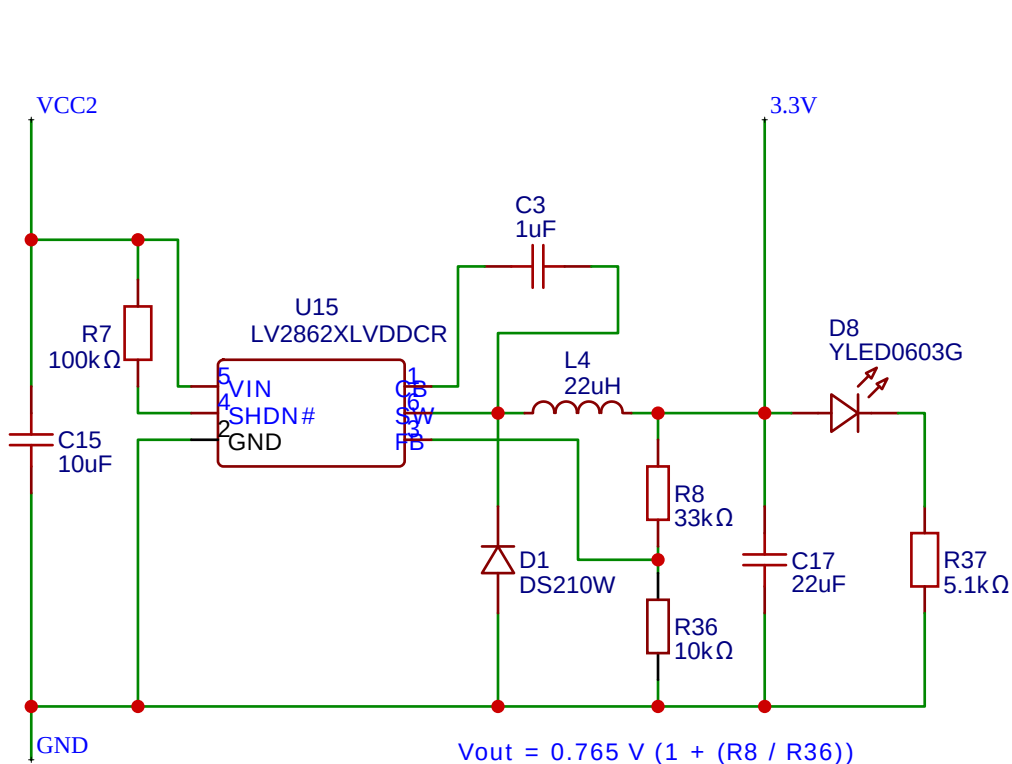
Buttons and rotary encoder



TFT

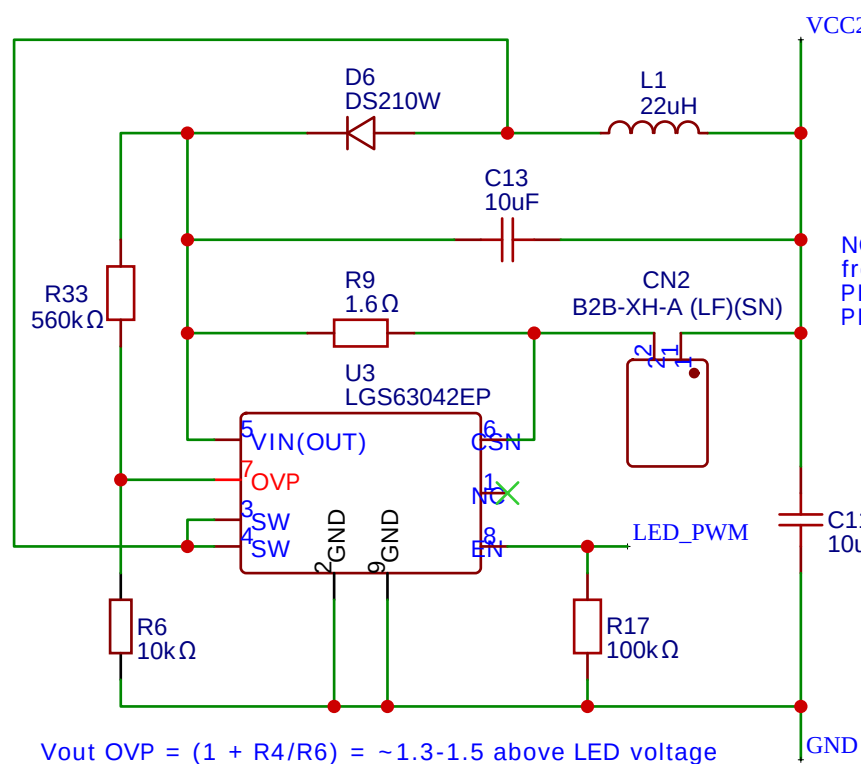


Power Supply 3.3V/0.6A

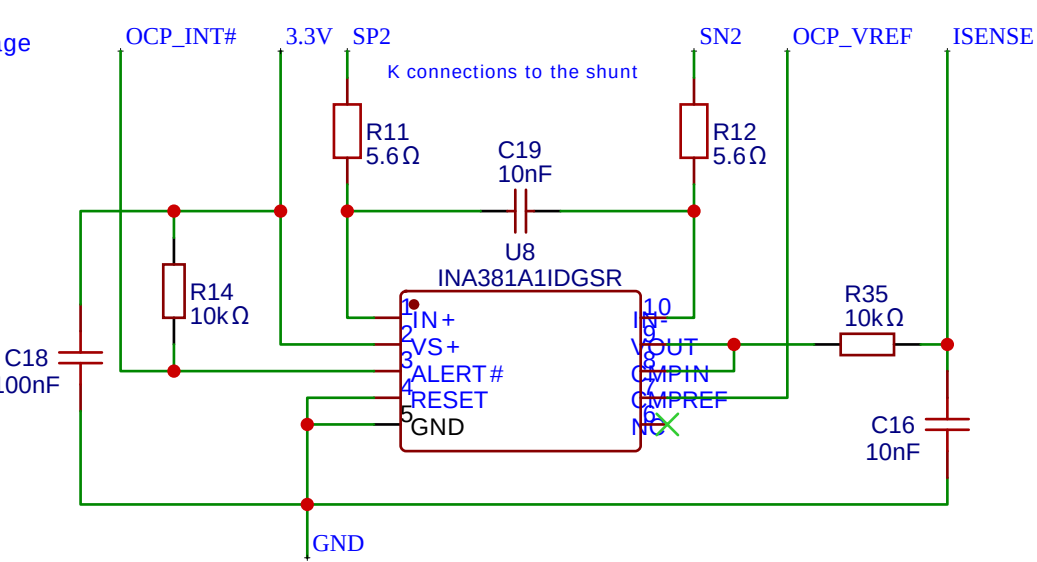


CC LED driver with PWM 2.5-32V/350mA

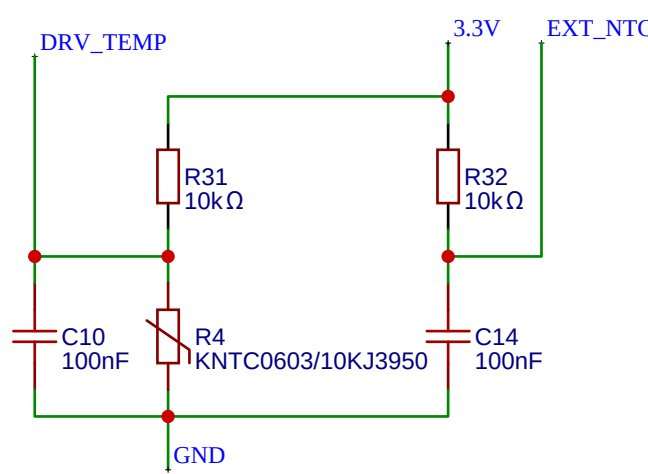
NOTE: If the LED voltage + VCC exceeds 57V, the OVP is triggered (OVP - VCC) / 1.3 = max. VLED, (57V - 36V) / 1.3 = ~16V



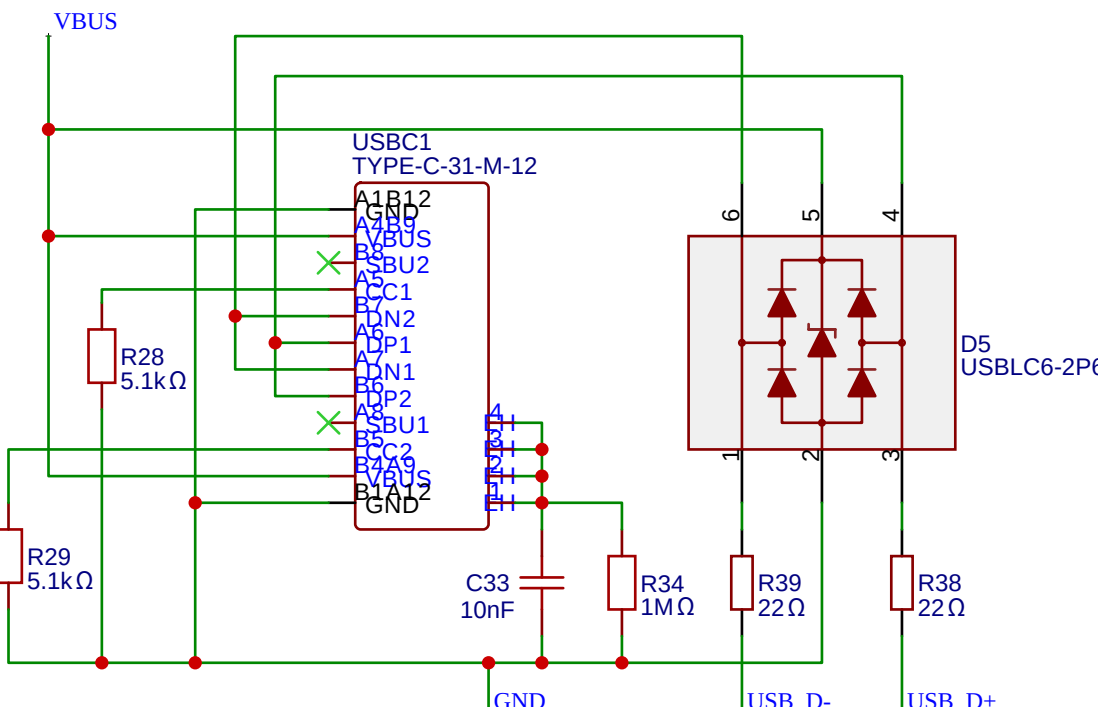
Input OCP



Temperature monitoring



USB connector for programming



TITLE: DC Motor Controller		REV: 1.1
Company: KFCLabs		Sheet: 1/1
Date: 2026-07-28		Drawn By: sascha23095123423